

Sheet 2 (Measurements and Errors)

- 1) The following set of ten measurements was recorded in the laboratory. Calculate the precision of the fourth measurement.

Measurement Number	Measurement Value X_n (volts)
1	98
2	102
3	101
4	97
5	100
6	103
7	98
8	106
9	107
10	99

Solution:

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^{10} x_i}{n} = \frac{98 + 102 + 101 + 97 + 100 + 103 + 98 + 106 + 107 + 99}{10} = 101.1$$

$$\text{precision} = 1 - \left| \frac{x_n - \bar{x}}{\bar{x}} \right| = 1 - \left| \frac{97 - 101.1}{101.1} \right| = 1 - 0.04 = 0.96$$

2. A set of voltmeter readings was taken by five observers as 99.7, 99.8, 100.0, 100.2, and 100.3 V. Calculate (a) the arithmetic mean of voltage. (b) the average deviation. (C) the standard deviation. (d) the variance.

Solution:

Voltmeter readings	d_i	$ d_i $	d_i^2
99.7	-0.3	0.3	0.3^2
99.8	-0.2	0.2	0.2^2
100	0	0	0
100.2	0.2	0.2	0.2^2
100.3	0.3	0.3	0.3^2
$\sum_{i=1}^5 x_i = 500$	$\sum_{i=1}^5 d_i = 0$	$\sum_{i=1}^5 d_i = 1$	

$$(a) \text{ Arithmetic Mean} = \bar{x} = \frac{\sum_{i=1}^5 x_i}{n} = \frac{500}{5} = 100$$

$$\text{Deviation } d_i = x_i - \bar{x}$$

$$(b) \text{ Average deviation} = d_{av} = \frac{\sum_{i=1}^5 |d_i|}{n} = \frac{0.3 + 0.2 + 0 + 0.2 + 0.3}{5} = \frac{1}{5} = 0.2$$

$$(c) \text{ Standard deviation} = \sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2}{n - 1}} \quad \text{where } n < 30$$

$$= \sqrt{\frac{0.3^2 + 0.2^2 + 0 + 0.2^2 + 0.3^2}{5 - 1}} = 0.2549$$

$$(d) \text{ Variance} = \sigma^2 = 0.2549^2 = 0.065$$

3. One hundred voltage readings were taken at small intervals of time and recorded to nearest 0.1 V. the frequency table is given below. Calculate (a) the arithmetic mean. (b) the average deviation. (C) the standard deviation. (d) the probable error.

Voltage reading (volt)	Frequency
99.7	2
99.8	8
99.9	20
100.0	40
100.1	21
100.2	6
100.3	3
	100

Solution:

Voltmeter readings	Frequency	d_i	$ d_i $	d_i^2
99.7	2	-0.3	0.3	0.3^2
99.8	8	-0.2	0.2	0.2^2
99.9	20	-0.1	0.1	0.1^2
100.0	40	0	0	0
100.1	21	0.1	0.1	0.1^2
100.2	6	0.2	0.2	0.2^2
100.3	3	0.3	0.3	0.3^2
	$\sum f_i = 100$			

$$(a) \text{ Arithmetic Mean} = \bar{x} = \frac{\sum_{i=1}^{100} x_i}{n}$$

$$= \frac{99.7 * 2 + 99.8 * 8 + 99.9 * 20 + 100 * 40 + 100.1 * 21 + 100.2 * 6 + 100.3 * 3}{100}$$

$$= 100$$

$$\text{Deviation } d_i = x_i - \bar{x}$$

$$\begin{aligned}
 \text{(b) Average deviation} &= d_{av} = \frac{\sum_{i=1}^{100} |d_i|}{n} \\
 &= \frac{0.3 \cdot 2 + 0.2 \cdot 8 + 0.1 \cdot 20 + 0 \cdot 40 + 0.1 \cdot 21 + 0.2 \cdot 6 + 0.3 \cdot 3}{100} = 0.084
 \end{aligned}$$

where $n \geq 30$. So,

$$\begin{aligned}
 \text{(c) Standard deviation} &= \sigma \\
 &= \sqrt{\frac{0.3^2 \cdot 2 + 0.2^2 \cdot 8 + 0.1^2 \cdot 20 + 0^2 \cdot 40 + 0.1^2 \cdot 21 + 0.2^2 \cdot 6 + 0.3^2 \cdot 3}{100}} \\
 &= 0.119
 \end{aligned}$$

$$\text{(d) probable error} = \pm 0.6745 \cdot \sigma = \pm 0.6745 \cdot 0.119 = \pm 0.0804 \text{ Volt}$$

4. A 500 V voltmeter is specified to be accurate within $\pm 1.5\%$ of full scale. Calculate the limiting error when the instrument is used to measure a 150 V source.

Solution:

$$\text{Max error } \Delta V = \pm \frac{1.5}{100} \cdot \text{full scale} = \pm \frac{1.5}{100} \cdot 500 = \pm 7.5 \text{ Volt}$$

$$\text{Limiting error\% of 150 V} = \frac{\Delta V}{V} = \pm \frac{7.5}{150} \cdot 100\% = \pm 5\%$$

5. A voltmeter and an ammeter are to be used to determine the power dissipated in a resistor. Both instruments are guaranteed to be accurate within $\pm 1\%$ at full scale deflection. If the voltmeter reads 100 V on its 150 V and the ammeter reads 55 mA on its 100 mA. Calculate the limiting error in the power in watt due to the previous voltmeter and ammeter readings.

Solution:

Accuracy = $\pm 1\%$ full scale deflection

$$\text{Voltage range} = 150 \text{ V}, \quad \Delta V = \frac{1}{100} \cdot 150 = 1.5 \text{ V}$$

$$\frac{\Delta V}{V} \% = \frac{1.5 \text{ V}}{100 \text{ V}} \cdot 100 = 1.5\%, \quad V = 100 \text{ V} \pm 1.5\%$$

$$\text{Ammeter range} = 100 \text{ mA}, \quad \Delta I = \frac{1}{100} \cdot 100 = 1 \text{ mA}$$

$$\frac{\Delta I}{I} \% = \frac{1 \text{ mA}}{55 \text{ mA}} \cdot 100 = 1.81\%, \quad I = 55 \text{ mA} \pm 1.81\%$$

$$P = VI = (100 \text{ V} \pm 1.5\%)(55 \text{ mA} \pm 1.81\%) = 5500 \text{ mW} \pm 3.31\%$$

$$\text{Limiting error in power } \Delta P = \pm \frac{3.31}{100} \cdot 5500 = \pm 182.05 \text{ mW}$$

6. A resistance was rated at 20 ohm and the measured current through this resistance was 8.2 A. the range of the ammeter was 10 A. Compute the power in watts dissipated in the resistor. The scale of ammeter had 100 divisions and could be read with certainty ± 0.5 division. It was later found that the resistance of the resistor was 0.2% greater than the specified resistance and the ammeter reading 1.0% more than the true value. Determine the known error in the computed power in watt and in percent of the computed power.

Solution:

$$R = R_{\text{true}} - 0.2\% = 20 \Omega - 0.2\%$$

$$\text{Ammeter range} = 10 \text{ A}$$

$$I = I_{\text{true}} + 1\%$$

$$I_{\text{true}} = 8.2 \text{ A} \pm 0.5 \text{ division} = 8.2 \text{ A} \pm \frac{0.5 \text{ div}}{100 \text{ div}} \cdot 10 \text{ A} = (8.2 \pm 0.05) \text{ A}$$

$$= 8.2 \pm \frac{0.05}{8.2} * 100\% = 8.2 \pm 0.61\%$$

$$I = I_{\text{true}} + 1\% = 8.2 \pm 0.61\% + 1\% = 8.2 + 0.61\% + 1\% = 8.2 \text{ A} + 1.61\%$$

$$P = I^2 R = (8.2 \text{ A} + 1.61\%)^2 (20 \Omega - 0.2\%) = 8.2^2 * 20 + (2 * 1.61\% - 0.2\%)$$

$$= 1344.8 \text{ W} + 3.02\%$$

$$\text{Limiting error in percent} \quad \frac{\Delta P}{P} \% = 3.02\%$$

$$\text{Limiting error in watt} \quad \Delta P = \frac{3.02}{100} * 1344.8 = 40.613 \text{ Watt}$$

7. The solution for the unknown resistance in a Wheatstone bridge is given by:

$$X = \frac{P}{Q} S$$

where $Q = 50 \pm 0.5\%$, $P = 100 \pm 0.5\%$, $S = 75.5 \pm 0.5\%$. Determine the magnitude of the unknown resistance X and the limiting error in ohm and in percent of X.

Solution:

$$X = \frac{P}{Q} S = \frac{(100 \pm 0.5\%)(75.5 \pm 0.5\%)}{50 \pm 0.5\%} = 151 \pm (0.5 + 0.5 - 0.5)\% = 151 \pm 0.5\%$$

$$\text{Magnitude of the unknown resistance } X = 151 \Omega.$$

$$\text{Limiting error in percent} \quad \frac{\Delta X}{X} \% = \pm 0.5\%$$

$$\text{Limiting error in ohm } \Delta X = \frac{\pm 0.5}{100} * 151 = \pm 0.75 \Omega$$

$$\mathbf{X} = (\quad) \, \Omega \pm \Delta \mathbf{X} \, \Omega$$

$$\mathbf{X} = (151 \pm 0.75) \, \Omega$$

$$\mathbf{X} = (\quad) \, \Omega \pm \frac{\Delta \mathbf{X}}{\mathbf{X}} \%$$

$$\mathbf{X} = 151 \pm 0.5\%$$

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